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Conclusion

The rearward amplification reducing controller provides a simple yet novel working controller with potential that can be used for autonomous driving LCVs on highways. It is shown that a linear single-track model could be used as a foundation for a controller with preliminary positive results. This was validated against a VTM model and real test data of an A-double.

The controller provides an intuitive and simple design that is easily tuned. It has the advantage of not requiring access to individual control actuators on all vehicle units or using sophisticated control methods. It only needs to know the vehicle combination it is driving combined with the parameters. It was shown that it could improve the generated rearward amplification ratio whilst not distorting the steering values and yaw-rates significantly. A low-pass filter and a Notch-filter was used as comparison and was tuned using the eigenfrequencies of the linear single-track model. For the highway manoeuvres the sanity checker outperformed the filters in the majority of the cases. The notch filter was shown to distort the results too much. A finer tuning of the filters is necessary to get a more qualitative measure against the controller.

The rearward amplification reducing controller could benefit from implementing a function which encourages the controller to consider safe driving behaviour when choosing between steering requests. Testing in simulation with an autonomous driving system, is needed before any closed loop tests are done on a real vehicle.

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A

Appendix 1

A.1 Derivation of single-track model for tractor and semitrailer

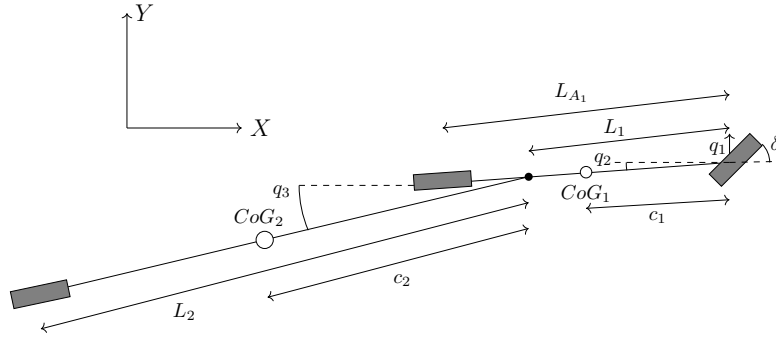


Figure A.1: Bicycle model of tractor and semitrailer.

Parameter	Symbol	Unit
Mass for the tractor	m_1	[kg]
Mass for the trailer	m_2	[kg]
Distance from front wheel to CoG for tractor	c_1	[m]
Distance from articulation point to CoG for trailer	c_2	[m]
Length from front wheels to articulation point	L_{A1}	[m]
Length from front axle to rear axle	L_1	[m]
Length from articulation point to rear axle of trailer	L_2	[m]
Constant longitudinal velocity	U	[m/s]
Cornering stiffness for wheel i	C_i	[N/°]
Inertia for unit j	I_j	[kgm ²]
Velocity	v	[m/s]
Generalized coordinate j	q_j	

Table A.1: Glossary for the variables used in derivation. Subscripts for the velocities, for example $Gx1$, should be read as: At CoG in x-axis for unit one.

Since planar motion is considered the potential energy $V=0$. Hence L only contains the kinematic energy. Looking at Figure A.1, the total kinematic energy from the tractor and semitrailer is

$$\begin{aligned}
T &= \frac{1}{2}m_1(v_{Gx1}^2 + v_{Gy1}^2) + \frac{1}{2}m_2(v_{Gx2}^2 + v_{Gy2}^2) + \frac{1}{2}I_1\dot{q}_2^2 + \frac{1}{2}I_2\dot{q}_3^2 \\
v_{Gx1}^2 &= \left(U + \dot{q}_2 c_1 \sin(q_2)\right)^2 \\
v_{Gy1}^2 &= \left(\dot{q}_1 - \dot{q}_2 c_1 \cos(q_2)\right)^2 \\
v_{Gx2}^2 &= \left(\dot{q}_3 c_2 \sin(q_3) + U + \dot{q}_2 L_{A_1} \sin q_2\right)^2 \\
v_{Gy2}^2 &= \left(-\dot{q}_3 c_2 \cos q_3 + \dot{q}_1 - \dot{q}_2 L_{A_1} \cos(q_2)\right)^2
\end{aligned} \tag{A.1}$$

Using small angle assumption and only keeping linear terms gives the velocities

$$v_{Gx1}^2 = \left(U + \dot{q}_2 c_1 \sin(q_2)\right)^2 \approx U^2 \tag{A.2}$$

$$v_{Gy1}^2 = \left(\dot{q}_1 - \dot{q}_2 c_1 \cos(q_2)\right)^2 \approx (\dot{q}_1 - \dot{q}_2 c_1)^2 \tag{A.3}$$

$$v_{Gx2}^2 = \left(\dot{q}_3 c_2 \sin(q_3) + U + \dot{q}_2 L_{A_1} \sin q_2\right)^2 \approx U^2 \tag{A.4}$$

$$v_{Gy2}^2 = \left(-\dot{q}_3 c_2 \cos q_3 + \dot{q}_1 - \dot{q}_2 L_{A_1} \cos(q_2)\right)^2 \approx (-\dot{q}_3 c_2 + \dot{q}_1 - \dot{q}_2 L_{A_1})^2 \tag{A.5}$$

Calculating the partial derivatives on the left hand side in equation 2.2 gives

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial v_{Gy1}^2}{\partial \dot{q}_1} \right) &= 2(\ddot{q}_1 - c_1 \ddot{q}_2) \\
\frac{d}{dt} \left(\frac{\partial v_{Gy2}^2}{\partial \dot{q}_1} \right) &= 2(\ddot{q}_1 - c_2 \ddot{q}_3 - \ddot{q}_2 L_{A_1}) \\
\frac{d}{dt} \left(\frac{\partial v_{Gy1}^2}{\partial \dot{q}_2} \right) &= -2c_1(\ddot{q}_1 - c_1 \ddot{q}_2) \\
\frac{d}{dt} \left(\frac{\partial v_{Gy2}^2}{\partial \dot{q}_2} \right) &= -2L_{A_1}(\ddot{q}_1 - \ddot{q}_2 L_{A_1} - \ddot{q}_3 c_2) \\
\frac{d}{dt} \left(\frac{\partial v_{Gy2}^2}{\partial \dot{q}_3} \right) &= -2c_2(\ddot{q}_1 - \ddot{q}_2 L_{A_1} - \ddot{q}_3 c_2) \\
\frac{d}{dt} \left(\frac{\partial v_{Gx1}^2}{\partial \dot{q}_1} \right) &= \frac{d}{dt} \left(\frac{\partial v_{Gx2}^2}{\partial \dot{q}_1} \right) = \frac{d}{dt} \left(\frac{\partial v_{Gx1}^2}{\partial \dot{q}_2} \right) = \frac{d}{dt} \left(\frac{\partial v_{Gx2}^2}{\partial \dot{q}_2} \right) = \dots \\
&= \frac{d}{dt} \left(\frac{\partial v_{Gx1}^2}{\partial \dot{q}_3} \right) = \frac{d}{dt} \left(\frac{\partial v_{Gy1}^2}{\partial \dot{q}_3} \right) = \frac{d}{dt} \left(\frac{\partial v_{Gx2}^2}{\partial \dot{q}_3} \right) = 0
\end{aligned} \tag{A.6}$$

Moreover, by checking the equations (A.1) one can easily observe that

$$\frac{\partial T}{\partial q_1} = \frac{\partial T}{\partial q_2} = \frac{\partial T}{\partial q_3} = 0. \tag{A.7}$$

Hence when adding (A.6) and (A.7) together with respect to each global coordinate the resulting system the equations become

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_1}\right) - \frac{\partial T}{\partial q_1} = \frac{1}{2}m_1\left(2(\ddot{q}_1 - \ddot{q}_2c_1)\right) + \frac{1}{2}m_2\left(2(-\ddot{q}_3c_2 + \ddot{q}_1 - \ddot{q}_2L_{A_1})\right) \quad (\text{A.8})$$

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_2}\right) - \frac{\partial T}{\partial q_2} &= \frac{1}{2}m_1\left(-2c_1(\ddot{q}_1 - c_1\ddot{q}_2)\right) \\ &\quad + \frac{1}{2}m_2\left(-2L_{A_1}(\ddot{q}_1 - \ddot{q}_2L_{A_1} - \ddot{q}_3c_2)\right) + 2\frac{1}{2}I_1\ddot{q}_2 \end{aligned} \quad (\text{A.9})$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_3}\right) - \frac{\partial T}{\partial q_3} = \frac{1}{2}m_2\left(-2c_2(\ddot{q}_1 - \ddot{q}_2L_{A_1} - \ddot{q}_3c_2)\right) + 2\frac{1}{2}I_2\ddot{q}_3 \quad (\text{A.10})$$

$$(\text{A.11})$$

Simplifying the equations yield

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_1}\right) - \frac{\partial T}{\partial q_1} = m_1(\ddot{q}_1 - \ddot{q}_2c_1) + m_2(-\ddot{q}_3c_2 + \ddot{q}_1 - \ddot{q}_2L_{A_1}) \quad (\text{A.12})$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_2}\right) - \frac{\partial T}{\partial q_2} = m_1c_1(-\ddot{q}_1 + c_1\ddot{q}_2) + m_2L_{A_1}(-\ddot{q}_1 + \ddot{q}_2L_{A_1} + \ddot{q}_3c_2) + I_1\ddot{q}_2 \quad (\text{A.13})$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_3}\right) - \frac{\partial T}{\partial q_3} = m_2c_2(-\ddot{q}_1 + \ddot{q}_2L_{A_1} + \ddot{q}_3c_2) + I_2\ddot{q}_3 \quad (\text{A.14})$$

Rearranging such that each variable is separated gives

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_1}\right) - \frac{\partial T}{\partial q_1} = \ddot{q}_1(m_1 + m_2) + \ddot{q}_2(-c_1m_1 - m_2L_{A_1}) + \ddot{q}_3(-c_2m_2) \quad (\text{A.15})$$

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_2}\right) - \frac{\partial T}{\partial q_2} &= \ddot{q}_1(-m_1c_1 - m_2L_{A_1}) + \ddot{q}_2(m_1c_1^2 + m_2L_{A_1}^2 + I_1) \\ &\quad + \ddot{q}_3(m_2c_2L_{A_1}) \end{aligned} \quad (\text{A.16})$$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_3}\right) - \frac{\partial T}{\partial q_3} = \ddot{q}_1(-m_2c_2) + \ddot{q}_2(m_2c_2L_{A_1}) + \ddot{q}_3(m_2c_2^2 + I_2) \quad (\text{A.17})$$

$$(\text{A.18})$$

The system of equations is then transformed into matrix form, gives the system becomes

$$\begin{pmatrix} m_1 + m_2 & -c_1m_1 - m_2L_{A_1} & -m_2c_2 \\ -m_1c_1 - m_2L_{A_1} & m_1c_1^2 + m_2L_{A_1}^2 + I_1 & m_2c_2L_{A_1} \\ -m_2c_2 & m_2c_2L_{A_1} & m_2c_2^2 + I_2 \end{pmatrix} \begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \\ \ddot{q}_3 \end{pmatrix} = \mathbf{Q} \quad (\text{A.19})$$

Now considering the right hand side of the equation. The total power done on the system is

$$P = P_1 + P_2 + P_3$$

where P_1, P_2 and P_3 is the power contributed from each of the wheels where the work on the wheel is calculated as:

$$P_1 = F_1(1, 0, 0) \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix}; \quad F_1 = C_1 \left(\delta + q_2 - \frac{v_{lat1}}{U} \right), \quad v_{lat1} = \dot{q}_1 \quad (\text{A.20})$$

$$P_2 = F_2(1, -L_1, 0) \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix}; \quad F_2 = C_2 \left(q_2 - \frac{v_{lat2}}{U} \right), \quad v_{lat2} = \dot{q}_1 - \dot{q}_2 L_1 \quad (\text{A.21})$$

$$P_3 = F_3(1, -L_{A1}, -L_2) \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix}; \quad F_3 = C_3 \left(q_3 - \frac{v_{lat3}}{U} \right), \quad (\text{A.22})$$

$$, v_{lat3} = \dot{q}_1 - \dot{q}_2 L_{A1} - \dot{q}_3 L_2$$

Since the power contribution w_1 can be written as

$$P_1 = F_1 \cdot v_1 \quad (\text{A.23})$$

and noticing that the work is a scalar product from a system of equations one can combine it with equation (2.3) to write the work on the following matrix form.

$$P = \mathbb{Q}^T \cdot \vec{q} = \begin{pmatrix} F_1 + F_2 + F_3 \\ -L_1 F_2 - L_{A1} F_3 \\ -L_2 F_3 \end{pmatrix}^T \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix} \quad (\text{A.24})$$

From (2.2) one notice that $\mathbb{Q}$ is the corresponds to the right hand side of the equation. Expanding $\mathbb{Q}$ gives

$$\mathbb{Q} = \begin{pmatrix} F_1 + F_2 + F_3 \\ -L_1 F_2 - L_{A1} F_3 \\ -L_2 F_3 \end{pmatrix} \quad (\text{A.25})$$

$$= \begin{pmatrix} C_1 \left(\delta + q_2 - \dot{q}_1 / U \right) + C_2 \left(q_2 - (\dot{q}_1 - \dot{q}_2 L_1) / U \right) + C_3 \left(q_3 - (\dot{q}_1 - \dot{q}_2 L_{A1} - \dot{q}_3 L_2) / U \right) \\ -L_1 C_2 \left(q_2 - (\dot{q}_1 - \dot{q}_2 L_1) / U \right) - L_{A1} C_3 \left(q_3 - (\dot{q}_1 - \dot{q}_2 L_{A1} - \dot{q}_3 L_2) / U \right) \\ -L_2 C_3 \left(q_3 - (\dot{q}_1 - \dot{q}_2 L_{A1} - \dot{q}_3 L_2) / U \right) \end{pmatrix}$$

Separating the equations into separate matrices gives

$$\mathbb{Q} = \frac{1}{U} \begin{pmatrix} -(C_1 + C_2 + C_3) & C_2 L_1 + C_3 L_{A1} & C_3 L_2 \\ L_1 C_2 + L_{A1} C_3 & -L_1^2 C_2 - L_{A1}^2 C_3 & -L_{A1} C_3 L_2 \\ L_2 C_3 & -L_{A1} L_2 C_3 & -L_2^2 C_3 \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{pmatrix} \quad (\text{A.26})$$

$$+ \begin{pmatrix} 0 & C_1 + C_2 & C_3 \\ 0 & -L_1 C_2 & -L_{A1} C_3 \\ 0 & 0 & -L_2 C_3 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} + \begin{pmatrix} C_1 \delta \\ 0 \\ 0 \end{pmatrix} \quad (\text{A.27})$$

Renaming the matrices gives

$$M = \begin{pmatrix} m_1 + m_2 & -c_1 m_1 - m_2 L_{A_1} & -m_2 c_2 \\ -m_1 c_1 - m_2 L_{A_1} & m_1 c_1^2 + m_2 L_{A_2}^2 + I_1 & m_2 c_2 L_{A_1} \\ -m_2 c_2 & m_2 c_2 L_{A_1} & m_2 c_2^2 + I_2 \end{pmatrix} = \begin{pmatrix} M_1 & M_2 & M_3 \\ M_4 & M_5 & M_6 \\ M_7 & M_8 & M_9 \end{pmatrix} \quad (\text{A.28})$$

$$N = \frac{1}{U} \begin{pmatrix} -(C_1 + C_2 + C_3) & C_2 L_1 + C_3 L_{A_1} & C_3 L_2 \\ L_1 C_2 + L_{A_1} C_3 & -L_1^2 C_2 - L_{A_1}^2 C_3 & -L_{A_1} C_3 L_2 \\ L_2 C_3 & -L_{A_1} L_2 C_3 & -L_2^2 C_3 \end{pmatrix} = \begin{pmatrix} N_1 & N_2 & N_3 \\ N_4 & N_5 & N_6 \\ N_7 & N_8 & N_9 \end{pmatrix} \quad (\text{A.29})$$

$$K = \begin{pmatrix} 0 & C_1 + C_2 & C_3 \\ 0 & -L_1 C_2 & -L_{A_1} C_3 \\ 0 & 0 & -L_2 C_3 \end{pmatrix} = \begin{pmatrix} K_1 & K_2 & K_3 \\ K_4 & K_5 & K_6 \\ K_7 & K_8 & K_9 \end{pmatrix} \quad (\text{A.30})$$

$$\Delta = \begin{pmatrix} C_1 \delta \\ 0 \\ 0 \end{pmatrix} \quad (\text{A.31})$$

By combining the left and right side terms one obtains the ODE-system

$$M\ddot{\vec{q}} - N\dot{\vec{q}} - K\vec{q} - \Delta = \mathbf{0}. \quad (\text{A.32})$$

Rearranging the equation such that $\ddot{\vec{q}}$ is isolated on the left hand side gives

$$\ddot{\vec{q}} = \underbrace{M^{-1}N}_{H}\dot{\vec{q}} + \underbrace{M^{-1}K}_{W}\vec{q} + \underbrace{M^{-1}\Delta}_{R} = H\dot{\vec{q}} + W\vec{q} + R, \quad (\text{A.33})$$

where $H, W \in \mathbb{R}^{3 \times 3}$ and $R \in \mathbb{R}^{3 \times 1}$. Introducing the new state vector x defined as

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix}, \quad x_i = \begin{cases} q_i, & i = 1, 2, 3 \\ \dot{q}_{i-3}, & i = 4, 5, 6 \end{cases} \quad (\text{A.34})$$

results the simpler linear ODE-system

$$\dot{\vec{x}} = A\vec{x} + B, \quad (\text{A.35})$$

where A and B are composed of

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ W_1 & W_2 & W_3 & H_1 & H_2 & H_3 \\ W_4 & W_5 & W_6 & H_4 & H_5 & H_6 \\ W_7 & W_8 & W_9 & H_7 & H_8 & H_9 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ R_1 \\ R_2 \\ R_3 \end{pmatrix}. \quad (\text{A.36})$$

A.2 Resulting rearward amplification from extreme case for different sizes of preview time and number of preview points

PT [ms] /PP	2	3	4	5	6	7	8	9	10	11
100	1.2796	1.2708	1.2775	1.2710	1.2710	1.2713	1.2728	1.2773	1.2780	1.2782
200	1.2889	1.2727	1.2714	1.2725	1.2730	1.2723	1.2634	1.2723	1.2723	1.2715
300	1.2792	1.2633	1.2463	1.2477	1.2451	1.2440	1.2456	1.2448	1.2477	1.2456
400	1.2897	1.2507	1.2489	1.2485	1.2493	1.2483	1.2489	1.2563	1.2692	1.2828
500	1.2353	1.2351	1.2353	1.2351	1.2363	1.2346	1.2345	1.2358	1.2736	1.2736
600	1.2258	1.2246	1.2261	1.2263	1.2258	1.2178	1.2178	1.2178	1.2178	1.2154
700	1.2262	1.1385	1.1510	1.1385	1.1538	1.2718	1.2720	1.2720	1.1525	1.1361
800	1.1355	1.1565	1.1502	1.1663	1.1290	1.1051	1.1290	1.1562	1.1537	1.1537
900	1.2060	1.2347	1.2347	1.2347	1.2378	1.2347	1.2347	1.2352	1.2352	1.2441
1000	1.1763	1.1786	1.1749	1.1755	1.1755	1.1755	1.1707	1.1707	1.2009	1.2009

Table A.2: Rearward amplification generated from different amounts of preview points and size of preview time distance. PP is the abbreviation of preview points and PT the preview time.

PT [ms] /PP	2	3	4	5	6	7	8	9	10	11
100	0.1759	0.1748	0.1758	0.1748	0.1749	0.1749	0.1751	0.1757	0.1759	0.1759
200	0.1765	0.1746	0.1745	0.1745	0.1746	0.1745	0.1734	0.1745	0.1745	0.1744
300	0.1760	0.1737	0.1710	0.1710	0.1709	0.1707	0.1708	0.1708	0.1711	0.1707
400	0.1759	0.1699	0.1698	0.1697	0.1698	0.1697	0.1698	0.1707	0.1717	0.1757
500	0.1683	0.1683	0.1683	0.1683	0.1683	0.1683	0.1682	0.1684	0.1737	0.1737
600	0.1614	0.1615	0.1615	0.1615	0.1614	0.1605	0.1605	0.1605	0.1605	0.1601
700	0.1597	0.1782	0.1725	0.1782	0.1724	0.1665	0.1665	0.1665	0.1794	0.1767
800	0.1885	0.1817	0.1822	0.1824	0.1759	0.1746	0.1759	0.1812	0.1819	0.1819
900	0.1464	0.1511	0.1511	0.1511	0.1515	0.1511	0.1511	0.1511	0.1511	0.1525
1000	0.1548	0.1502	0.1503	0.1502	0.1502	0.1502	0.1503	0.1503	0.1549	0.1549

Table A.3: Maximum absolute yaw-rate in rad/s observed from different amounts of preview points and size of preview time distance. PP is the abbreviation of preview points and PT the preview time.